Deep Learning

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Notes Repository

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1 Introduction

In this section, we will briefly introduce the basic concepts of deep learning and its historical background.

1.1 Basic Definitions

AI Artificial Intelligence (AI) is a discipline that has the ambition to create machines that mimic human intelligence. It encompasses a wide range of techniques and approaches, including rule-based systems, expert systems, and machine learning. The ultimate goal of AI is to develop systems capable of learning from experience as effectively as humans do.

Machine Learning Machine Learning (ML) is a subset of AI that enables machines to learn from experience (i.e., data) to improve their performance on a specific task.

Deep Learning This is a subfield of Machine Learning that particularly focuses on **Artificial Neural Networks** (ANNs). The term 'deep' refers to the use of multiple layers in the network, which allows it to learn complex representations of data, with multiple levels of abstraction.

1.2 Historical Background

The concept of artificial neural networks dates back to the 1940s, with the work of Warren McCulloch and Walter Pitts, who proposed a simple model that attempted to emulate the behavior of biological neurons. This is regarded as the first single-layer perceptron model, which is a fundamental building block of deep learning.

Electronic Brain In the first perceptron model, the inputs form a vector of **binary features** that are simply summed together and passed through a **threshold function**, without any **weights** involved.

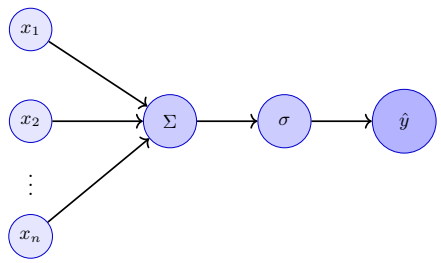

Figure 1: Architecture of the electronic brain model. Inputs x_i are summed together and passed through an activation function σ to produce the output $\hat{y}$.

It can be shown that this model is capable of classifying linearly separable data, such as the AND, OR, and NOT functions, but it fails to classify non-linearly separable data, such as the XOR function.

The Perceptron In the 1950s, Frank Rosenblatt introduced the **Perceptron**, which is an extension of the electronic brain model. The key difference is the introduction of **weights** for each input, which allows the model to learn from data and adjust its parameters accordingly.

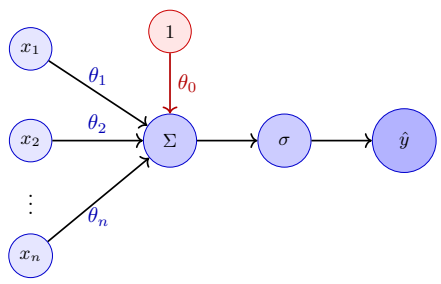

Figure 2: Architecture of a single layer perceptron. Inputs x_i are weighted by w_i, summed together, and passed through an activation function σ to produce the output y.

The prediction $\hat{y}$ of the perceptron is given by the following formula:

$$\hat{y} = \sigma\left(\sum_{i=1}^{n} \theta_i x_i + \theta_0\right) \tag{1.1}$$

where θ_i are the weights associated with each input x_i, and σ is the activation function. We can also express the prediction in vector form as follows:

$$\hat{y} = \sigma\left(\boldsymbol{\theta}^T \boldsymbol{x} + \theta_0\right) \tag{1.2}$$

The most important intuition is that the weights θ_i are not fixed, but rather learned from data.

The bias term is introduced to allow the model to shift the decision boundary (hyperplane) away from the origin.

Perceptron Learning Algorithm The weights of the perceptron are learned from data using the **Perceptron Training Rule**. Given a set of training examples $\mathcal{T} = \{(\boldsymbol{x_1}, y_1), (\boldsymbol{x_2}, y_2), \ldots, (\boldsymbol{x_N}, y_N)\}$, $y_i \in \{0, 1\}$ and a learning rate η, the weights are updated as follows:

1. Initialize the weights θ_i to small random values;
2. Until convergence, repeat:
   (a) Select randomly a training example $(\boldsymbol{x_i}, y_i)$ from $\mathcal{T}$;
   (b) Compute the prediction $\hat{y}_i$ using the current weights:
   $$\hat{y}_i = \sigma\left(\boldsymbol{\theta}^T \boldsymbol{x_i} + \theta_0\right)$$
   (c) Update the weights based on the error between the prediction and the true label:
   $$\boldsymbol{\theta}^{(t+1)} \leftarrow \boldsymbol{\theta}^{(t)} + \eta(y_i - \hat{y}_i)\boldsymbol{x_i}$$
   (d) Update the bias:
   $$\theta_0^{(t+1)} \leftarrow \theta_0^{(t)} + \eta(y_i - \hat{y}_i)$$

Note that we are assuming that the training data is linearly separable, otherwise we would have to introduce a stopping criterion to prevent the algorithm from running indefinitely.

Perceptron Today Today, the activation function σ is typically a non-linear function, such as the sigmoid, ReLU, or tanh function, which allows the model to learn complex patterns in the data.

Perceptron

An artificial neuron, i.e., a perceptron is a non-linear parametrized function with a restricted output range.

Perceptron Problem As anticipated, in the 1970s it was found that the perceptron was not able to learn non-linearly separable data, such as the XOR function. This was seen as a major setback for the field of AI, and it led to a decline in interest and funding for research in this area, which is now known as the "AI winter".

MLP AI did not stop there, and in the 1980s, researchers introduced the concept of **Multilayer Perceptron** (MLP). This architecture had a major issue, which was the difficulty in computing the weights of the hidden layers.

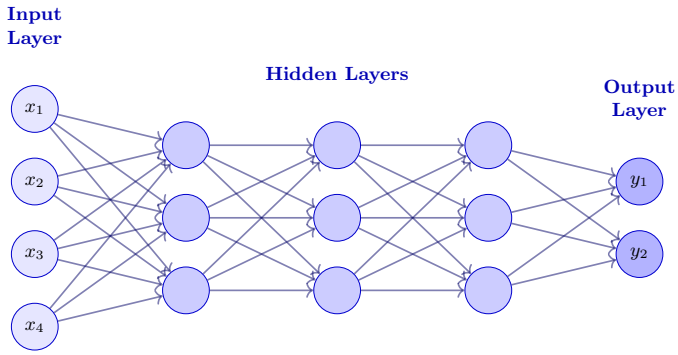

Figure 3: Architecture of a Multi-Layer Perceptron (MLP) with three hidden layers, fully connected in a feed-forward fashion.

This structure was very flexible as multiple nodes could be used in the output layer, so it could be used for multi-class classification problems, and also for regression.

Backpropagation The perceptron learning algorithm was not suitable for training MLPs, as no expected output was available for the hidden layers, but just for the output layer.

Backpropagation is an efficient algorithm for computing the gradients also for large datasets. At a high level, the algorithm can be summarized as follows:

1. **Forward propagation**: Given an input $\boldsymbol{x}$, compute the output of the network $\hat{y}$ by propagating the input through the layers of the network;
2. **Loss computation**: Compute the loss $L(y, f(\boldsymbol{x}, \boldsymbol{\theta}))$, where y is the true label and $f(\boldsymbol{x}, \boldsymbol{\theta})$ is the output of the network with parameters $\boldsymbol{\theta}$;
3. **Backward propagation**: the error signal is propagated backwards through the network and it's used to update the weights.

Obviosly, this concept will be covered in more detail in the next chapters.

Other Architectures Other architectures were proposed in the following years that lead to the current breakthrough of deep learning, such as:

- **Convolutional Neural Networks** (CNNs), which are particularly effective for image processing tasks;

- **Recurrent Neural Networks** (RNNs), used for sequential data, such as time series and natural language processing;

Cybenko's Theorem Cybenko's theorem states that a 2-layer feed-forward neural network with linear output can approximate any continuous function on a compact subset of $\mathbb{R}^n$ with arbitrary accuracy.

In practice, the more hidden layers we have, the more complex functions we can learn. Deep neural networks need a lot of labeled data to be trained, and cannot be trained easily.

Second Winter of AI Three main factors contributed to the second AI winter in the late 1980s:

- **Deep Neural Networks** tend to overfit the training data, and had problems with vanishing gradients, which made them difficult to train;
- **Lack of computational resources** to train deep neural networks. In fact, no GPUs were available at the time;
- **Lack of large datasets** to train deep neural networks.

All these issues led to a decline in interest and funding for research in this area, with an increase of interest in other areas of ML, such as Support Vector Machines (SVMs) and Kernel Methods.

First Deep Learning Applications - AlexNet In 2012, a deep convolutional neural network called AlexNet won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) with a top-5 error rate of 15.3%, which was significantly better than the second-best entry, which had an error rate of 26.2%. This was a major breakthrough for the field of deep learning, and it led to a shift of interest and funding back to this area, which is now known as the "Deep Learning Revolution".

AlexNet also shifted the focus of research from shallow architectures to deep architectures. The first one required heavy works on feature engineering before training the model, while the second one was able to learn features directly from the data, without any human intervention.

2 Machine Learning Recap

Machine Learning is a subset of AI, born in contrast to the programming paradigm, which is based on the idea of writing explicit rules to solve a problem.

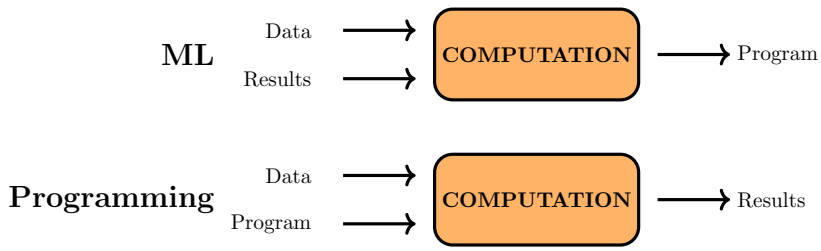

Figure 4: Programming paradigm vs Machine Learning paradigm.

In the Machine Learning paradigm, we are instead given the data, along with the results, and we try to learn the **model**. The learned model (e.g. splits in a decision tree, linear function for SVMs) represents the solution to the problem, and it can be used to make predictions on new data.

2.1 Why do we need Machine Learning

The main reason why we need Machine Learning can be summarized in the following points:

- **Complexity of the problem**: For many tasks, we have no human expertise to write explicit rules;
- **Personalized solutions**: For many tasks, we need to learn personalized solutions for each user, which is not feasible with the programming paradigm;
- **Huge amount of data**: Working with huge amount of data is not feasible, as it would require writing down explicit rules for each possible combination of inputs.

2.2 Machine Learning - Another definition

Machine Learning

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at

tasks in T, as measured by P, improves with experience E.

This definition is more formal and precise than the previous one, and it highlights the importance of the three components: experience, task, and performance measure. Each machine learning algorithm can be characterized by these three components. For example, taking in consideration a spam email filter, we can define:

- T: Categorizing emails as spam or not spam;
- P: Accuracy of the classification. Other metrics may be more appropriate;
- E: Database of emails labeled as spam or not spam by human experts.

2.3 Types of learning

We can identify three main types of learning:

- **Supervised learning**: We are given a training set of input-output pairs, and we want to learn a function that maps inputs to outputs.
- **Unsupervised learning**: We are given a training set of inputs, and we want to learn a function that captures the underlying structure of the data.
- **Reinforcement learning**: We are given an environment, and we want to learn a policy that maximizes the cumulative reward.

2.3.1 Supervised Learning

Supervised learning can be further divided into subcategories, such as:

- **Classification**: The output is a discrete label. More formally, given a training set $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ we can define it as:

$$f : \mathcal{X} \to \{1, \ldots, k\} \tag{2.1}$$

  where $\mathcal{X}$ is the input space, and $\{1, \ldots, k\}$ is the set of possible labels.

  Based on the number of classes, we can further divide classification into **binary classification** and **multi-class classification**.

- **Regression**: The output is a continuous value. More formally, given a training set $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ we can define it as:

$$f : \mathcal{X} \to \mathbb{R} \tag{2.2}$$

- **Structured output**: **supervised learning** doesn't necessarily require a **vector** as input, and a **scalar** as output. We can have more complex input and output structures, such as images, graphs, or sequences. In all cases the output is a structured object, with relationships between its components. Some practical example of **structured input regression** problems are:
  - **Semantic segmentation**: Given an image of input size $M \times N \times 3$, we want to predict a mask of size $M \times N$, where each pixel is labeled with a class. The original solution to this kind of problem was to use **feature engineering** to extract features from neighborhoods of pixels. After that, SVMs were used to make predictions based on the extracted features. **Neural Networks** show their real power in this kind of problems, as they are able to learn features directly from the data, without any human intervention.
  - **Machine translation**: given a sentence (sequence of words) in a source language, we want to translate it to a target language. This is even more complex than the previous example, as not only the output is related to its adjacent components, but also grammatical and semantic relationships between words need to be captured.

2.3.2 Unsupervised Learning

In unsupervised learning, we are given a training set of inputs, without any corresponding output labels, and we want to learn the underlying structure of the data. Some examples of unsupervised learning tasks are:

- **Clustering**: grouping similar data points together;
- **Deep clustering**: a technique that combines deep learning with classical clustering algorithms, such as k-means, to learn better representations of the data and improve clustering performance. It is based on the idea of using a deep neural network to learn a low-dimensional representation of the data. This is then used as input to a clustering algorithm, producing a set of **pseudo-labels** that are used to train the neural network using backpropagation. This process is repeated iteratively until convergence;
- **Generative Models**: models learn the underlying distribution of the data, and can be used to generate new data points, similar to the ones in the training set. Also in this case, we can have **deep generative models**. They try to map exaples extracted from a simple distribution Z to a more complex distribution $g_\theta(Z)$, which is similar to the distribution of X.
- **Dimensionality reduction**: try to map data to a lower-dimensional space in which visualization and interpretation are easier. This is often used as a preprocessing step for other machine learning algorithms.

- **Autoencoders**: a type of encoder-decoder architecture, where:
  - The **encoder** maps the input data to a **latent space** of lower dimensionality;
  - The **decoder** maps the latent representation back to the original input.

  The two components are trained together to minimize the reconstruction error, which is the difference between the original input and the reconstructed output. Once trained, the encoder can be used alone to obtain lower-dimensional representations of the data.

2.3.3 Reinforcement

3 Feed Forward Neural Networks